ANA 221 Angiology

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Angiology is the study of body vasculature.

- All arteries carry blood away from the heart while all veins take blood to the heart.
- All arteries carry oxygenated blood except pulmonary artery that carry de-oxygenated blood.
- All vein carry de-oxygenated blood except pulmonary vein that carry oxygenated blood.
- The body vasculature include; arteries, veins and lymphatic vessels.
- Each play specific role.

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The artery: All arteries are pulsatile because of the pressure pump from the heart beats.

- Arterial walls are usually much thicker than its venous counterparts to withstand the high pressure of blood flow from the heart.
- The arterial wall is tough on the outside while smooth inside for easy flow of blood.
- Artery contain strong muscular middle part that aid blood pump.
- Arteries have no valves except at the beginning of the large arteries i.e. ascending aorta or pulmonary trunk.
- Ascending aorta arising from left ventricle and pulmonary trunk from right ventricle are the largest arteries.
- Where two arteries meet is called anastomosis.

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The veins: There is no blood pump in the veins.

- Venous wall is thin in contrast to arterial wall that is thick.
- Veins return blood back to the heart.
- Veins have valves to ensure blood flows in one direction i.e. against gravity but some veins are without valves.
- The presence of valves in superficial veins of arm was a major factor of Harvey's discovery of circulation of blood.
- Veins without valves include: (i) Superior and inferior vena cava that communicates with right atrium.
- (ii) Ophthalmic vein (iii) Dural sinuses (iv) Vertebral vein (v) Cardiac veins (vi) Pulmonary veins (vii) hepatic vein (viii) Pelvic vein (ix) Portal vein.

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- The weakening of venous valve leads to a clinical condition known as varicose veins.
- It occurs when faulty valves allow blood to flow in the wrong direction making the veins to become enlarged.
- It happens in the veins near the surface (superficial veins) and it often appears bluish.



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- There are two types of anastomosis; Actual and Potential.
- Actual: Where artery meet end to end i.e. arteries of curvatures of stomach.
- Potential: By terminal arterioles; It takes time for it to dilate to take sufficient blood
- Capillaries: The arteries end in capillaries. It is the capillaries that connect the artery to the vein although in some situations there is arteriovenous blood shunt i.e. abnormal connections between coronary arteries.
- Artery deliver the oxygenated blood to the capillaries where the actual exchange of oxygen and carbon dioxide occurs.
- The capillary then deliver the waste-rich blood to the veins for export back to the heart and lungs.
- Capillaries do not have valves.

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Variations:

- Vascular pattern has wide variations. Like in embryology, the fate of vessels supplying an organ that migrates far from its original appearance, in one case the vessel may persist by merely elongating to reach the organ as in the case of gonads. In another situation series of new vessels may appear replacing the older ones, like in the case of kidney.

- It develops in pelvic region where it is supplied by median sacral artery but following ascension to lumbar region the supply is replaced by lumbar arteries.

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